

A Spectral Interpretation of Quantum Measurement via Boundary-Induced Diagonalization

Abstract

We present a structural reinterpretation of quantum measurement as a boundary-induced spectral diagonalization process. While the formal decoherence framework is well established, we show that quantum collapse can be understood as a spectral stabilization of structurally privileged modes induced by boundary interactions. This interpretation provides a unified spectral–ontological bridge between quantum measurement, decoherence, and linear spectral resolution.

1. Introduction

In standard quantum mechanics, measurement is described as a projection of the system state onto an eigenbasis of an observable. Decoherence theory explains the suppression of off-diagonal terms of the reduced density matrix under environmental interaction, yet the ontological status of the resulting pointer states remains debated.

In this work, we show that quantum measurement can be formally interpreted as a spectral diagonalization process induced by boundary interaction, structurally analogous to the resolution of linear systems via normal equations.

2. Density matrix and boundary interaction

Let a quantum system S interact with a boundary B . The joint state evolves as

$$|\Psi\rangle = \sum_i c_i |s_i\rangle |b_i\rangle.$$

Tracing over the boundary degrees of freedom yields the reduced density matrix

$$\rho_S = \text{Tr}_B(|\Psi\rangle\langle\Psi|).$$

Environmental interaction induces decay of off-diagonal terms in a preferred basis.

3. Lemma 1: Hermiticity and positivity

The reduced density matrix ρ_S is Hermitian and positive semidefinite:

$$\rho_S^\dagger = \rho_S, \quad \langle v | \rho_S | v \rangle \geq 0.$$

If the system is not in a pure eigenstate, ρ_S admits a nontrivial spectral decomposition.

4. Lemma 2: Spectral decomposition

By the spectral theorem,

$$\rho_S = \sum_i p_i |\phi_i\rangle \langle \phi_i|,$$

with $p_i \geq 0$ and $\sum_i p_i = 1$.

The vectors $|\phi_i\rangle$ define the spectral basis selected by boundary interaction.

5. Spectral stabilization

Decoherence drives the density matrix toward its diagonal spectral form. This process selects stable spectral modes and suppresses incoherent superpositions.

Thus, quantum evolution under boundary interaction is equivalent to a spectral stabilization process.

6. Structural interpretation

The eigenvectors $|\phi_i\rangle$ represent structurally privileged quantum directions. The system is not arbitrarily projected, but resolved along these spectral directions.

The coefficients p_i quantify the structural weights of these modes.

7. Spectral–ontological interpretation

We interpret each eigenvector $|\phi_i\rangle$ as a **quantum delta**, representing a stable ontological configuration under boundary interaction.

Measurement does not create the delta; it reveals the delta already stabilized by spectral diagonalization.

8. Theorem (Spectral Measurement Theorem)

Theorem.

Let ρ_S be the reduced density matrix of a quantum system under boundary interaction. Then quantum measurement outcomes correspond to the eigenvectors of ρ_S , and the measurement probabilities are given by the associated eigenvalues.

Thus, quantum measurement is a spectral resolution of boundary-induced deltas.

9. Conceptual novelty

Although spectral decomposition of density matrices is standard, its interpretation as a boundary-induced collapse into ontological deltas has not been previously formalized.

This interpretation provides a structural bridge between quantum mechanics and linear spectral resolution.

10. Relation to linear systems

The decomposition

$$\rho_S = \sum_i p_i |\phi_i\rangle \langle \phi_i|$$

plays the same formal role as

$$A^T A = Q D Q^T$$

in linear system resolution. In both cases, a boundary-induced operator is diagonalized to reveal stable spectral modes.

11. Relation to Boundary-Induced Collapse (BIC)

The diagonalization of ρ_S corresponds to a particular realization of Boundary-Induced Collapse, where the boundary is the environment and the operator is the reduced density matrix.

12. Conclusion

Quantum measurement is not an independent postulate but a structural consequence of boundary-induced spectral stabilization. The emergence of quantum deltas is mathematically inevitable in any decohering quantum system.

13. Closing sentence

Quantum measurement, like linear resolution, is a spectral necessity induced by boundary interaction.

14. Disclaimer

The formalism of density matrix diagonalization is standard; the novelty lies exclusively in its structural and ontological interpretation.